

Lambda Function

A **lambda function** is a **small anonymous (nameless) function** used to perform a simple operation in a single line.

It is mainly used when you need a function **only once**.

Why Lambda?

Suppose we need the function only once.

Instead of writing four lines,

```
def square(x):
```

```
    return x * x
```

Python lets you write it in **one line and** only difference is that the lambda version is shorter.

```
square = lambda x: x * x
```

```
print(square(5))
```

Output

25

This is called a **lambda function**.

Syntax

lambda arguments : expression

Parts

lambda

Keyword used to create a lambda function.

arguments

Input values (like parameters).

:

Separates the arguments from the expression.

expression

The operation to perform.

Examples

Largest Number

largest = lambda a, b: a if a > b else b

```
print(largest(10, 20))
```

Output

20

Even or Odd

even = lambda x: x % 2 == 0

```
print(even(8))
```

```
print(even(7))
```

Output

True

False

Why is it called an Anonymous Function?

Normally,

```
def add():
```

The function has a **name** (add).

But in lambda,

```
lambda x: x*x
```

The function is created **without a name**, so it is called an **anonymous function**.

When we write

```
square = lambda x: x*x
```

the variable square simply stores the anonymous function.

When Should We Use Lambda?

Use lambda when:

- The function is very small.
 - It contains only one expression.
 - You need the function only once.
 - You want shorter code.
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When Shouldn't We Use Lambda?

Don't use lambda if:

- The logic is long.
- There are multiple statements.
- You need loops or several if statements.
- The function will be reused many times.

In these cases, use a normal function with def.

Real-Life Example

Suppose we have marks:

```
marks = [45, 80, 60, 90]
```

We want to add 5 marks to each student.

```
new_marks = list(map(lambda x: x + 5, marks))
```

```
print(new_marks)
```

Output

```
[50, 85, 65, 95]
```

Here, the lambda function adds 5 to each element.

Lambda vs Normal Function

S.No	Normal Function	Lambda Function
1.	Uses def keyword	Uses lambda keyword
2.	Has a function name	Anonymous (no name)
3.	Can contain multiple statements	Only one expression
4.	Uses return	No return keyword (returns automatically)
5.	Best for complex logic	Best for simple operations
